INCIDENT LIFE CYCLE AUTOMATION

Technology Stack & Application Characteristics

Technology choices kept intentionally focused on the project scope

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DOCUMENT DATE **08 October 2026**	SCOPE **Incident lifecycle workflow & automation only**

Component	Purpose	Proposed Technology
User Interface	Incident entry, detail, timeline and dashboard	HTML + CSS + JavaScript
Application Logic	Validation, lifecycle rules, routing and status transitions	Python + Flask
Automation	SLA checks and escalation processing	Python scheduler / background worker
Database	Incident and lifecycle data	SQLite for project prototype
Notifications	Assignment and escalation messages	SMTP / email service
API Layer	Communication between interface and backend	Flask routes / JSON endpoints
Runtime	Local execution or suitable deployment environment	Python runtime

Application Characteristics

- Layered separation between interface, business rules, data and automation.
- Role-based access for actions that change incident state.
- Audit logging for key lifecycle changes.
- Background SLA checks should not block normal incident operations.
- The stack is proposed for the academic implementation and should match the actual build before final submission.